CoWiC-MAC: Project Technical Document

Project: CoWiC-MAC Case Study
Document: `reviews/project_document.md`
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Base paper: Cheng, T., Li, S., and Pan, Y. "CoWiC-MAC: A Coordinated WiFi-Centric MAC Mechanism in Heterogeneous Wireless Networks." *Computer Networks*, Vol. 287, Art. 112479, Sept. 2026. DOI: 10.1016/j.comnet.2026.112479

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1. Problem Statement

WiFi (IEEE 802.11) and ZigBee (IEEE 802.15.4) share the 2.4 GHz ISM band. They use different MAC protocols. They cannot negotiate channel access with each other.

Uncoordinated operation causes Cross-Technology Interference (CTI). The base paper reports ZigBee packet loss above 60% and WiFi throughput loss near 20% under heavy shared-channel load.

CoWiC-MAC solves this with a WiFi-centric coordination loop:

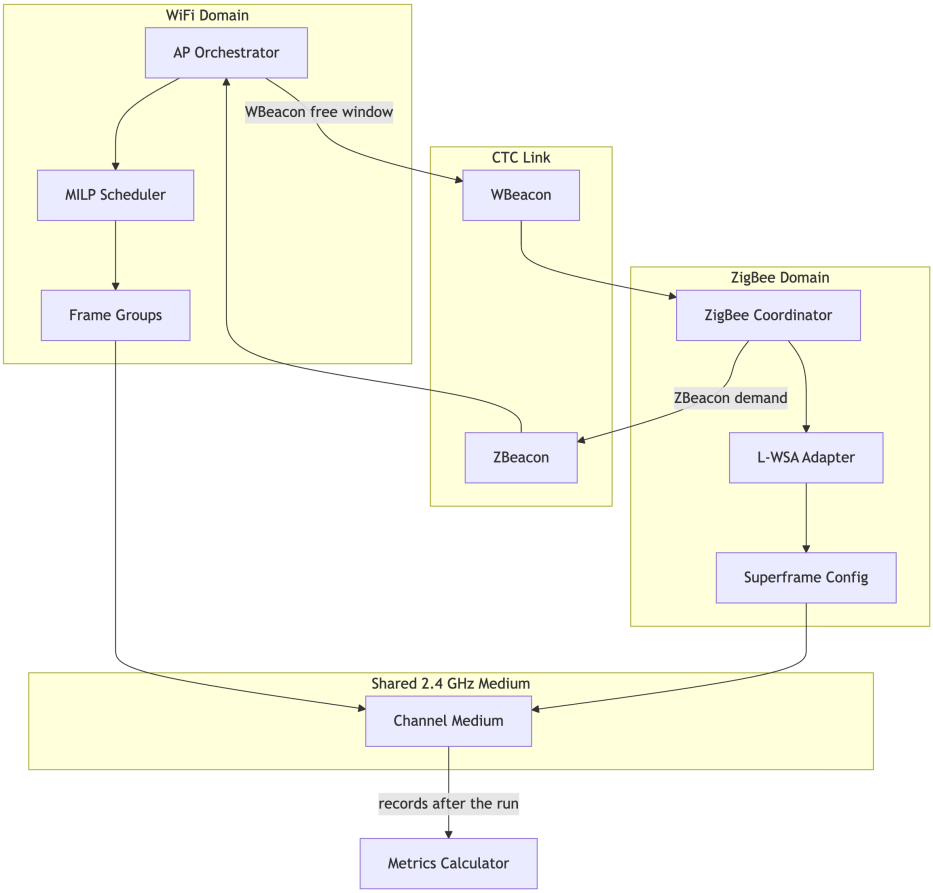
1. The WiFi Access Point (AP) reschedules Frame Groups (FGs) with Mixed-Integer Linear Programming (MILP).
2. The ZigBee Coordinator (ZC) adapts its superframe with the Lightweight Window-Scaling Algorithm (L-WSA).
3. The two networks exchange control messages over Cross-Technology Communication (CTC).

This project builds a software-only simulation of that loop. It uses no radio hardware.

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2. System Architecture

The system has three domains: the WiFi domain, the ZigBee domain, and the shared 2.4 GHz medium.



End-to-end coordination cycle

1. The ZC reports demand with a ZBeacon (`d_free_ms`).
2. The AP solves the MILP and creates a free window plus a sensing window.
3. The AP sends a WBeacon with window bounds.
4. The ZC runs L-WSA and builds a superframe that fits the free window.
5. Both sides transmit on the shared medium.
6. The channel medium records each transmission and marks overlaps as collisions. The metrics module reads these records after the run.

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3. WiFi Domain

3.1 Frame Groups

A Frame Group (FG) is a batch of WiFi frames that the AP schedules as one unit. This matches the 802.11ax Target Wake Time (TWT) and trigger-frame model in the base paper.

Main fields:

Field	Meaning
`fg_id`	Unique identifier
`arrival_time_ms`	Earliest time the FG may start
`duration_ms`	Airtime duration
`deadline_ms`	Latest finish time
`original_start_ms`	Start time before rescheduling
`qos_sensitive`	Flag for delay-sensitive traffic

3.2 MILP Scheduler (Equations 11-17)

The AP solves this programme each cycle:

$$\min \sum_{k=1}^{K} z_k^{\text{free}} d_k + \gamma \sum_{k=1}^{K} z_k^{\text{sense}} d_k + \lambda \sum_{k=1}^{K} y_k + \sum_{k=1}^{K} \mu_k P_k$$

Decision variables:

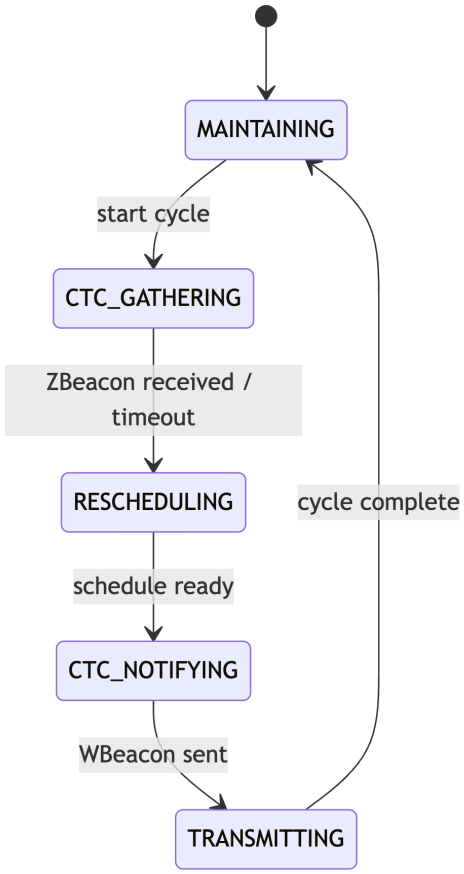
- s'_k — scheduled start time of FG k
- y_k — delay relative to the original start
- $z_k^{\text{free}}, z_k^{\text{sense}}$ — overlap indicators with free and sensing windows
- P_k — postpone binary (defer into the next cycle)
- b_{kj} — ordering binary between FG pairs

Constraint groups:

1. Delay definition: $y_k \geq s'_k - s_k$, $y_k \geq 0$
2. Causality: $a_k \leq s'_k + MP_k$
3. Deadline and cycle bound: $s'_k + d_k \leq e_k + MP_k$ and $s'_k + d_k \leq T_{\text{end}} + MP_k$
4. Feasibility of postponed FGs: $T_{\text{end}} + d_k \leq e_k + M(1 - P_k)$
5. Inter-FG non-overlap via Big-M ordering binaries b_{kj}
6. Overlap indicator bounds for z_k^{free} and z_k^{sense}
7. Sensing window protection: $\sum_{k=1}^{K} z_k^{\text{sense}} = 0$

Solver: PuLP with the CBC solver that PuLP includes. A HiGHS solver can be supplied as an option. If the MILP has no optimal solution, the AP uses whitespace expansion fallback.

3.3 AP Orchestrator State Machine



States:

State	Action
`MAINTAINING`	Idle between cycles
`CTC_GATHERING`	Collect ZigBee demand
`RESCHEDULING`	Run MILP or fallback
`CTC_NOTIFYING`	Send WBeacon
`TRANSMITTING`	Drive WiFi airtime

3.4 Baseline Experiments (Exp 1-3)

Exp	Name	Behaviour
Exp 1	Uncoordinated	No CTC. No rescheduling. ZigBee uses a fixed superframe at the cycle start. Carrier sensing is not modelled.
Exp 2	Opportunistic	AP senses natural idle gaps. No WiFi rescheduling.
Exp 3	ECC	NAV-based reservation. Fixed ZigBee superframe.

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4. ZigBee Domain

4.1 IEEE 802.15.4 Superframe

A beacon-enabled superframe has:

- 1 beacon slot
- i Contention Access Period (CAP) slots
- j Contention Free Period (CFP) slots

Hard limits from the standard and the base paper:

- $0 \leq j \leq 7$
- $i + j \leq 15$
- Total active slots: $1 + i + j$

IEEE 802.15.4 2.4 GHz O-QPSK PHY constants used in this project:

Constant	Value
Symbol duration	0.016 ms
Base slot duration	0.96 ms (60 symbols)
Backoff slot	0.32 ms (20 symbols)
Turnaround time	0.192 ms (12 symbols)
ACK duration	0.352 ms (22 symbols)
Max packet duration	4.064 ms (254 symbols)

4.2 L-WSA Algorithm (Algorithm 2)

The ZC maximises:

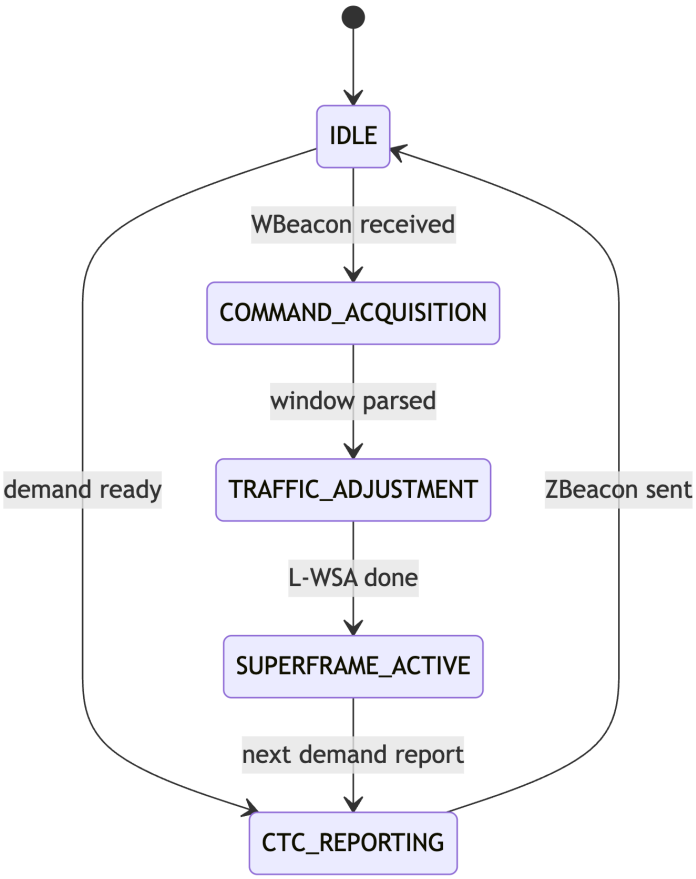
$$\eta_{\text{total}} = m \cdot t_{\min} \cdot \eta_{\text{cap}}^{p} + \alpha \cdot n \cdot t_{\text{SfSlot}} \cdot \eta_{\text{cfp}}^{p}$$

Search variables:

Variable	Domain
t_{SfSlot}	`BASE_SLOT * 2^SO`, $SO \in [0, 8]$
j (CFP slots)	$[0, 7]$
i (CAP slots)	$i + j \leq 15$
t_{offset}	Window start, or end of each residual WiFi fragment inside the window

Filter: $(1 + i + j) \cdot t_{\text{SfSlot}} \leq$ free-window duration. Each candidate slot must avoid residual WiFi fragments $\mathcal{T}^{\text{WiFi}}$.

4.3 ZigBee Coordinator State Machine



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5. Cross-Technology Communication

CTC lets WiFi and ZigBee exchange control messages without a shared MAC.

Message	Direction	Payload keys
ZBeacon	ZC → AP	`d_free_ms`
WBeacon	AP → ZC	`t_free_start_ms`, `t_free_end_ms`, `t_sense_start_ms`, `t_sense_end_ms`

Physical-layer CTC primitives in the base paper:

- **WiCast** — WiFi → ZigBee downlink by payload emulation
- **SymBee** — ZigBee → WiFi uplink by phase-shift symbol patterns

This software model abstracts CTC as discrete-event messages with a configurable packet error rate (PER). It does not emulate RF waveforms.

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6. Shared Channel Model

`ChannelMedium` records every WiFi and ZigBee transmission interval. A collision occurs when intervals overlap beyond `OVERLAP_TOLERANCE_MS`.

The medium stores one record per transmission: protocol, start time, end time, and a success flag.

The metrics module computes airtime, delay, jitter, and drops after the run. It uses these records and the WiFi scheduling decisions of the engine.

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7. Experiments

Exp	Config	What it tests	Fixed parts
Exp 1	`exp1_uncoordinated.yaml`	Uncoordinated baseline	No CTC, no MILP, no L-WSA
Exp 2	`exp2_opportunistic.yaml`	Natural whitespace only	No WiFi rescheduling
Exp 3	`exp3_ecc.yaml`	NAV-style ECC	Fixed ZigBee superframe
Exp 4	`exp4_cowic_full.yaml`	Full CoWiC-MAC	Soft MILP + L-WSA + sensing window
Exp 5	`exp5_rigid_vacate.yaml`	Hard-constraint vacate	Hard MILP + fixed ZigBee superframe
Exp 6	`exp6_overhead_free.yaml`	CoWiC without sensing overhead	Soft MILP + L-WSA, no sensing window

Parameter sweep

Parameter	Values
D_{free}	31, 62, 93, 123 ms
T_{ZB}	1000, 2000, 5000 ms
Traffic profiles	Traffic I and III by default; I–IV with `--traffic`
D_{sensing}	10, 20, 50, 100 ms
Seeds	5 per point

Acceptance criterion (Issue #11): Exp 4 at $D_{\text{free}} = 31$ ms and $T_{ZB} = 1000$ ms must achieve $G_z > 0.9$ and $PDR_w = 0$.

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8. Key Metrics

Metric	Definition
G_z	ZigBee delivered payload / theoretical clean-spectrum capacity
PDR_w	Discarded WiFi FG duration / total requested WiFi duration
U_{eff}	Successful WiFi + ZigBee + control airtime / simulated duration
Jitter	Variation of WiFi FG delay
Max delay	Largest WiFi FG delay in the run

Hypotheses under test

1. Under heavy load, CoWiC-MAC achieves $G_z \geq 90\%$.
2. CoWiC-MAC keeps $PDR_w = 0$ by deadline-aware postponement.
3. The ablations show which parts are necessary under fragmented whitespace. Exp 5 removes L-WSA (fixed superframe with hard WiFi vacate). Exp 6 removes the sensing window.

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9. Software Stack

Layer	Choice
Language	Python 3.11+
DES engine	SimPy
MILP	PuLP + CBC (`pulp < 3.0.0`)
Data	NumPy, Pandas, PyYAML
Plots	Matplotlib
Tests	pytest (245 tests)
Lint / types	ruff, mypy

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10. Presentation Ownership

Presenter	Domain
Upayan Mazumder	Central architecture
Nitesh Pandey	WiFi side
Divyansh Saxena	ZigBee side

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